

# FinCopilot

## Feature Ideas for Future Development

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The current build already covers RAG over SEC filings, a sentiment-augmented LSTM price predictor, a scheduler for daily predictions, watchlists, and Alpaca paper trading with portfolio tracking. The features below are grouped by how directly they build on that existing infrastructure.

### 1. Natural Extensions

*Build directly on tables and services that already exist.*

#### **Filing comparisons**

Answer questions like “How did AAPL's risk factors change year over year?” by diffing filing\_chunks across filing dates already stored in the database.

#### **Multi-company / peer queries**

Ask about several tickers at once and get a side-by-side answer, since companies and filings are already keyed by ticker.

#### **Alerts on watchlist**

Send an email or push notification when a watched ticker's daily prediction flips, or price moves more than a set threshold, using the existing watchlists and predictions tables plus the daily scheduler.

#### **Prediction confidence and calibration view**

Show a calibration curve (predicted probability vs. actual outcome) instead of just directional accuracy, reinforcing the project's emphasis on honest metrics.

#### **Portfolio analytics**

Compute Sharpe ratio, max drawdown, win rate per trade, and sector exposure from the existing transactions and portfolio\_snapshots tables.

### 2. Complementary New Features

*New capabilities that plug into the existing pipelines.*

#### **Source-highlighted answers**

Show the exact filing excerpt(s) the RAG answer drew from, with a link to the section — reinforces the faithfulness evaluation already tracked in eval/.

#### **Earnings call transcripts**

Add transcripts as a new ingestion source alongside 10-Ks and 10-Qs, reusing the same chunk-and-embed pipeline.

#### **News sentiment timeline**

Plot the FinBERT sentiment score over time next to price so users can visually inspect the feature the LSTM is using.

#### **Explain-the-trade**

When the paper trader executes a trade, generate a short LLM explanation citing the prediction and sentiment that triggered it — ties the ML and RAG halves of the project together.

#### **Leaderboard / model versioning**

Track multiple model versions (baseline vs. sentiment-augmented LSTM) side by side with live accuracy, surfacing the ablation already documented in the README.

### **3. More Ambitious Additions**

*Larger features that extend the product surface.*

#### **Backtesting sandbox UI**

Let a user pick a date range and see how the paper strategy would have performed, using the existing `backtest_daily.py` logic behind a UI.

#### **Custom alerts via natural language**

Parse requests like “notify me if NVDA's sentiment turns negative” into a structured alert rule.

#### **Social / shareable predictions**

A public, read-only page per ticker showing the current signal and accuracy track record — useful as a shareable portfolio-piece link.